

- Arithmetic Operators -

Arithmetic operators are the symbols Python uses to perform mathematical calculations on numbers.

Python has seven arithmetic operators.

1. Addition ($\pm$)

Add the value on the left of the operator to the value on the right.

eg. $p = 12$

$q = 4$

$result = p \pm q$

Print(result)

$result = left + right$

$12 + 4 = 16$

$\rightarrow 16$

2. Subtraction (-)

Subtract the value on the right from the value

on the left

$$p = 12 \quad (12) - (4) = 8$$

$$q = 4$$

$$\text{result} = p - q$$

Print (result) $\rightarrow 8$

left $\rightarrow$ right

$$(15) - (10) = 5$$

3. Multiplication (*)

multiplies the two values together.

$$\text{eg. } p = 12$$

$$q = 4$$

$$\text{result} = p * q$$

$$\text{print (result)} \rightarrow 48$$

4. Division (/)

Divides the left value by the right value & always returns a float.

eg. $p = 12$
 $q = 4 \rightarrow$
 $result = p / q$
 $Print(result) \rightarrow \underline{3.0}$

Print
 $\textcircled{3} \rightarrow \text{float}$
 $\begin{array}{r} 4 \overline{) 12} \\ \underline{12} \\ 0 \end{array}$

5. Floor Division (//)

Divides the left value by the right value & it always returns integer.

$$\text{eg. } p = 12$$

$$q = 4$$

$$\text{result} = p // q$$

$$\text{Print (result)} \rightarrow \underline{\underline{3}}$$

$$\begin{array}{r} 12 \overline{) 4} \\ \underline{3} \\ 12 \\ \underline{12} \\ 00 \end{array}$$

6. Modulus (%)

Returns only the remainder that is left over after dividing the left value by the input value.

eg. $P = 12$
 $Q = 4$

$$\text{result} = P \% Q$$

$$\text{Print}(\text{result}) \rightarrow 0$$

$$\begin{array}{r} 3 \\ 4 \overline{) 12} \\ \underline{12} \\ 0 \end{array}$$

eg. $P = 13$

$$Q = 4$$

$$\text{result} = P \% Q$$

$$\text{Print}(\text{result})$$

$$\text{o/p} = 1$$

$$\begin{array}{r} 3 \\ 4 \overline{) 13} \\ \underline{12} \\ 01 \end{array}$$

7. Exponentiation (**)

Raise the left value to the power of the right value.

Q.
 base = 4
 power = 3
 result = base ^{power}
 Print (result) → 64

$$4^3 = \frac{4 \times 4 \times 4}{16 \times 4}$$

$$\frac{\cancel{3 \times 3 \times 3 \times 3}^2}{\cancel{9 \times 3}^7 \times 3}$$

$$\frac{\cancel{4 \times 4 \times 4}^3}{4^3} = \frac{2 \times 16}{\times 4} = \underline{\underline{64}}$$

$$\text{result} + 12 = \underline{2} * * \underline{3}$$

$$\text{result} + 13 = \underline{4} * \cancel{4}$$

$$4 \times 4 \times 4 \times 4$$

$$\underline{16 \times 4}$$

$$64 \times 4$$

$$\underline{\underline{\text{result} + 13 = 256}}$$

$$\underline{2} \times \underline{2} \times \underline{2}$$

$$4 \times 2$$

$$= \underline{8}$$

$$\begin{array}{r} 1 \\ 64 \\ \times 4 \\ \hline 256 \end{array}$$

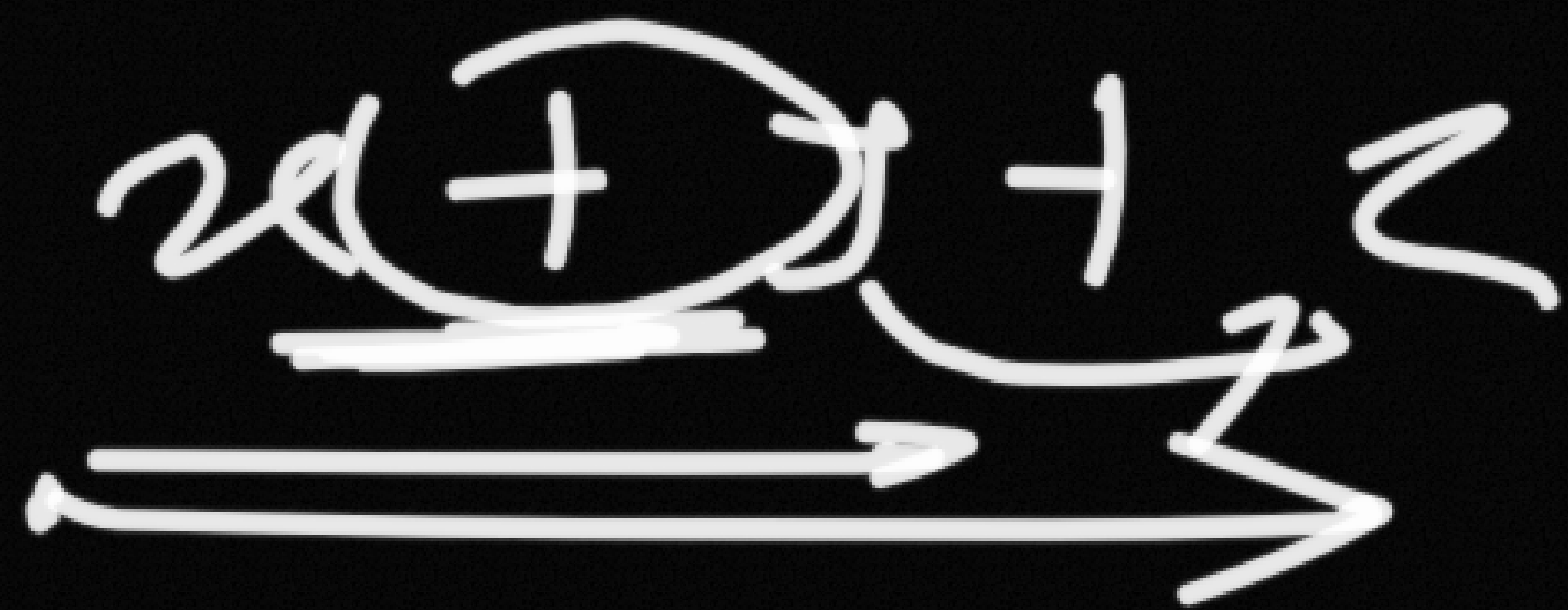
Precedence (Order of operations)

- When an expression contains more than one operator, Python has to decide which operation to carry out first.

eg.

$\boxed{9 + 3} * 2$ $12 * 2$ $= \underline{\underline{24}}$	$9 + \boxed{3 * 2}$ $9 + 6$ $= \underline{\underline{15}}$
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- operators with higher precedence are evaluated before operators with lower precedence,
- if two operators share the same precedence, python evaluates them left to right.



1. Parentheses ()
2. Exponentiation (~~*~~ **)
3. Multiplication, division, floor division, modulus (*, /, //, %)
4. Addition & Subtraction (+, -)

1. Parentheses () - Anything written inside round brackets is always calculated first

$$\begin{aligned}\text{eg. value} &= \underline{(9+3)} * 2 \\ &= 12 * 2 \\ &= 24\end{aligned}$$

2. Exponentiation ($*$ $*$ $*$) $2 \times 2 \times 2$
Exponentiation is calculated before
mul, div, add, sub. 6 $*$ 3

$$\begin{aligned} \text{eg. Value} &= 4 + 2 \times \times 3 \\ &= 4 + 8 \\ &= 12 \end{aligned}$$

3. mul, div, floor div, modulus -

(*, /, //, %) - Same precedence level.

which is lower than () & **

but higher than + & -.

if same precedence - left to right.

$$\begin{aligned} \text{eg. value} &= 18 - 3 * 4 \\ &= 18 - 12 \\ &= 6 \end{aligned}$$

4. Addition & Subtraction
Same precedence level - left to
right evaluate.

Q. Value $\pm$ $\overset{\text{left}}{20 - 5} + \overset{\text{right}}{3}$

$$\begin{array}{r} \underline{20 - 5} + 3 \\ 15 + 3 \\ 18 \end{array}$$

